Crashworthiness Evaluation of 3D-Printed PLA+ and PLA-CF Tubes with Varying Geometries Using Additive Manufacturing

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PROBLEM STATEMENT,CRITERIA AND CONSTRAINTS

Problem Statement:
- Need for lightweight structures in automotive and packaging.
- Limitations of traditional heavy metallic crash boxes.
- Exploration of 3D-printed polymer tubes as energy absorbers.

Evaluation Criteria:
- Specific Energy Absorption (SEA)
- Peak Crush Force (PCF)
- Mean Crush Force (MCF)
- Crush Force Efficiency (CFE)

Constraints:
- Material availability
- 3D printing limitations
- Safety during testing

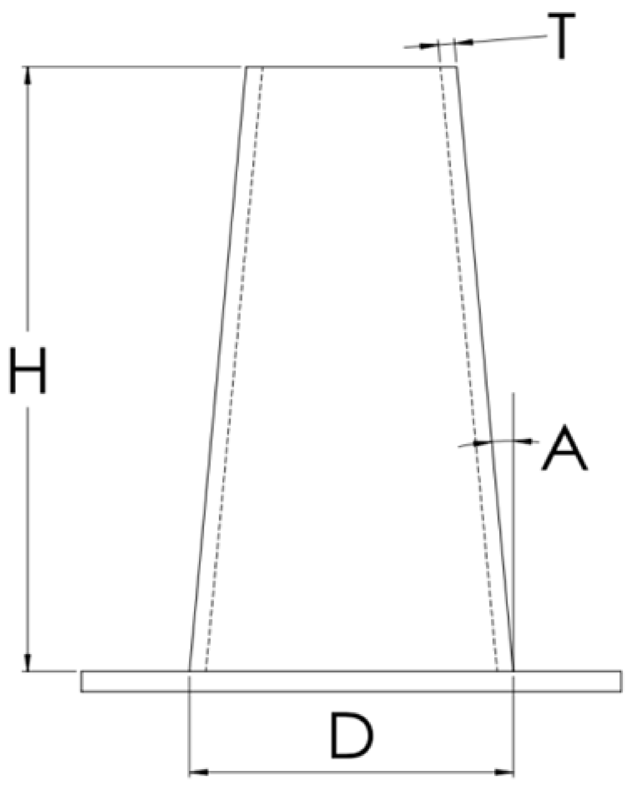
Limitation

Existing Solutions:
- Metallic structures (high strength, heavy, costly).
- Composite materials (complex, expensive manufacturing).

Limitations:
- Metallic structures
- weight
- manufacturing complexity
- cost

Research Gap:
- Insufficient data on crashworthiness of 3D-printed polymers (PLA+, PLA-CF) with varied geometries.

PROPOSED SOLUTION & DESIGN



PLA-CF

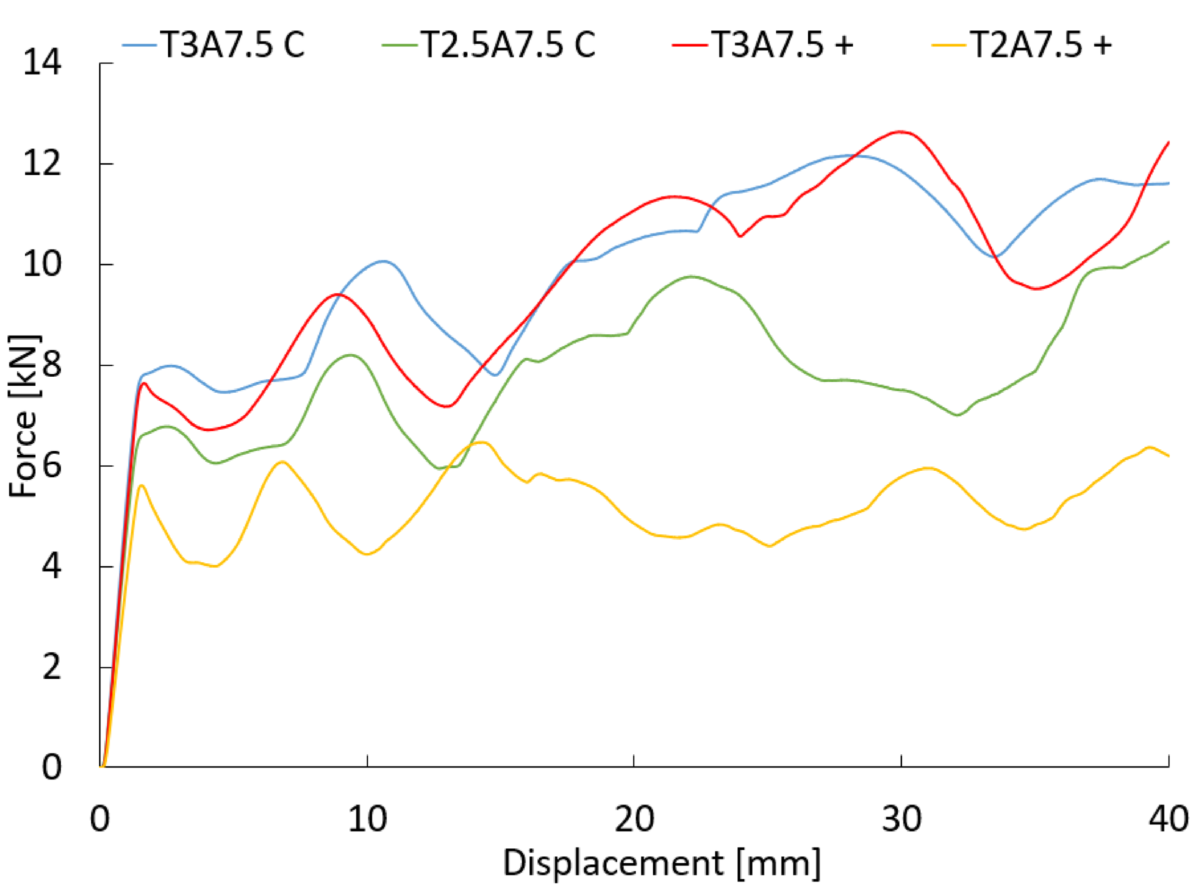
PLA+

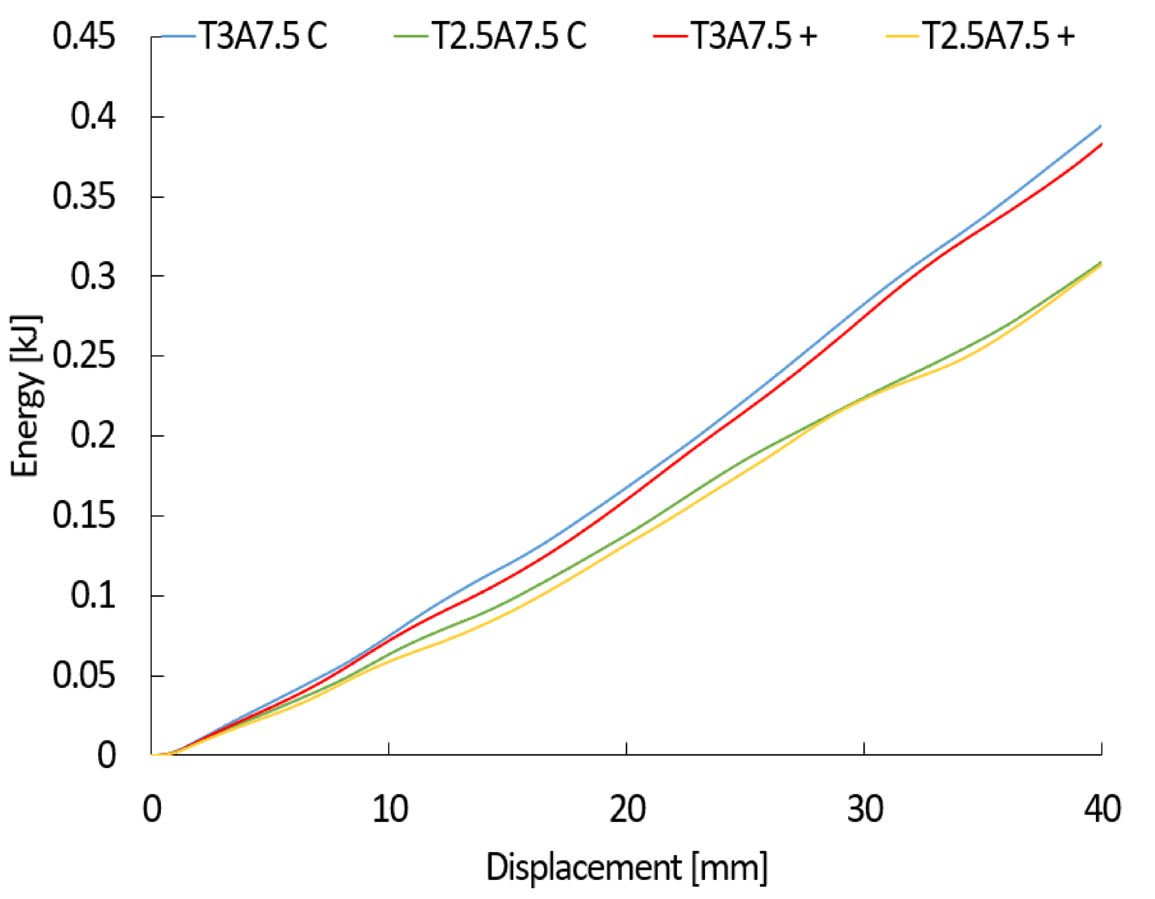
PLA-CF

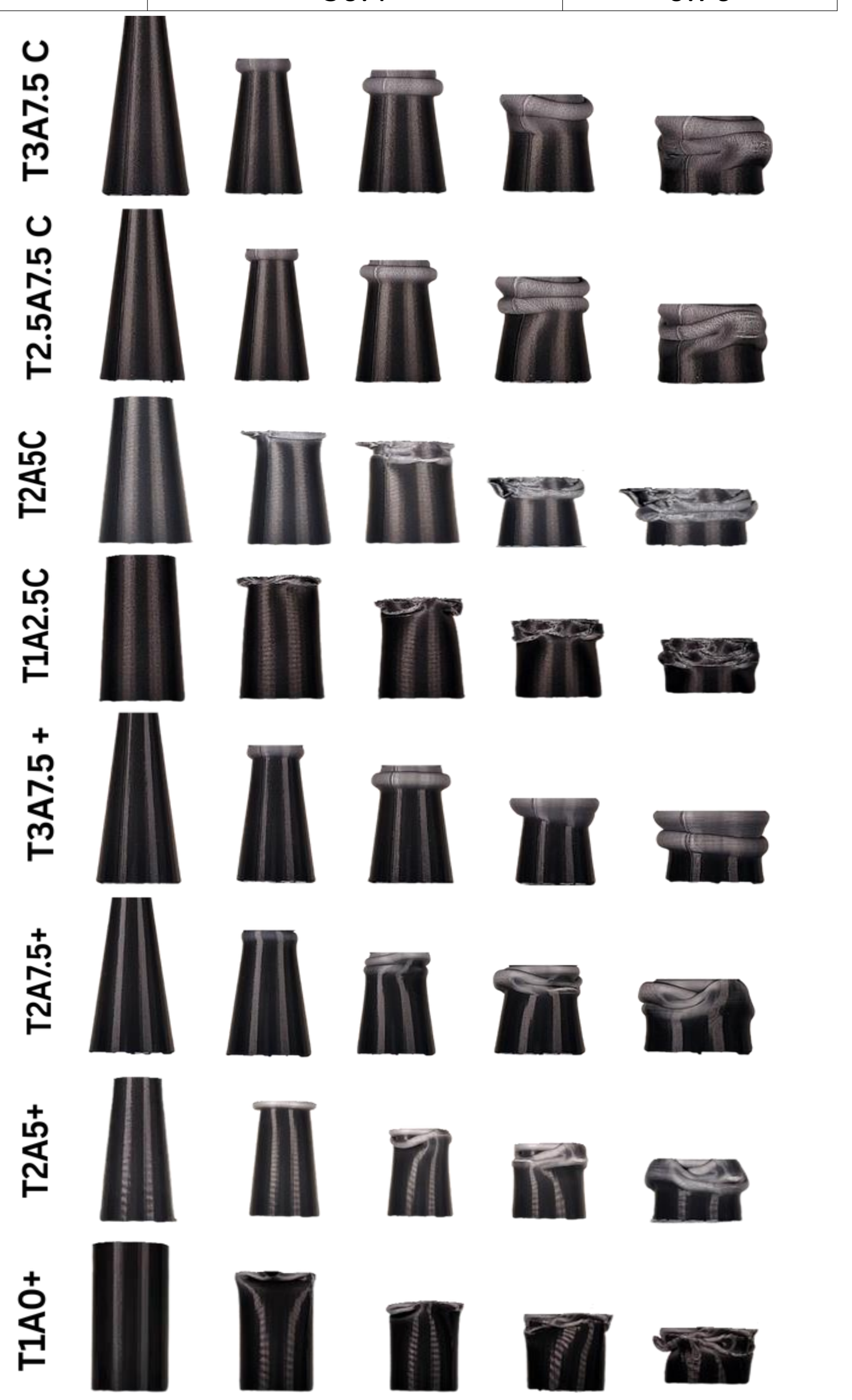
PLA+

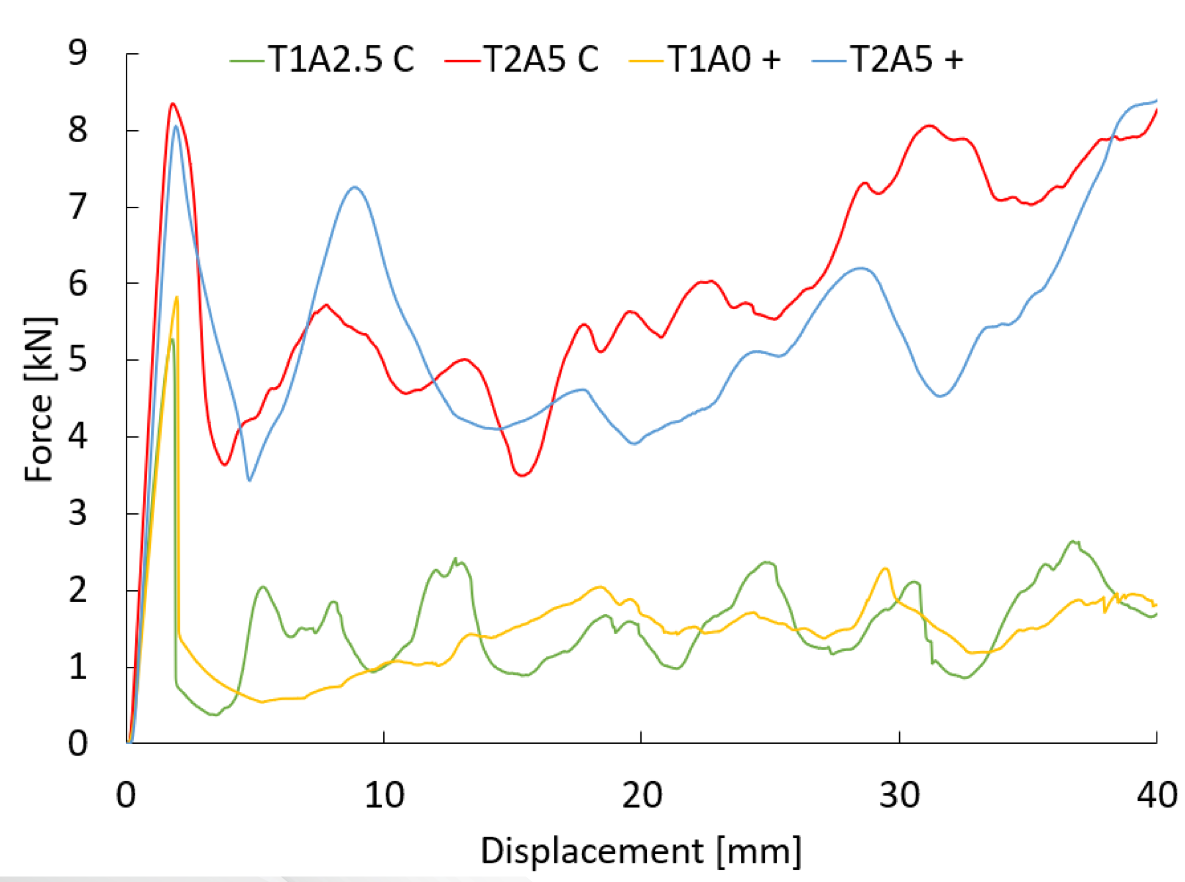
IMPLEMENTATION AND RESULTS

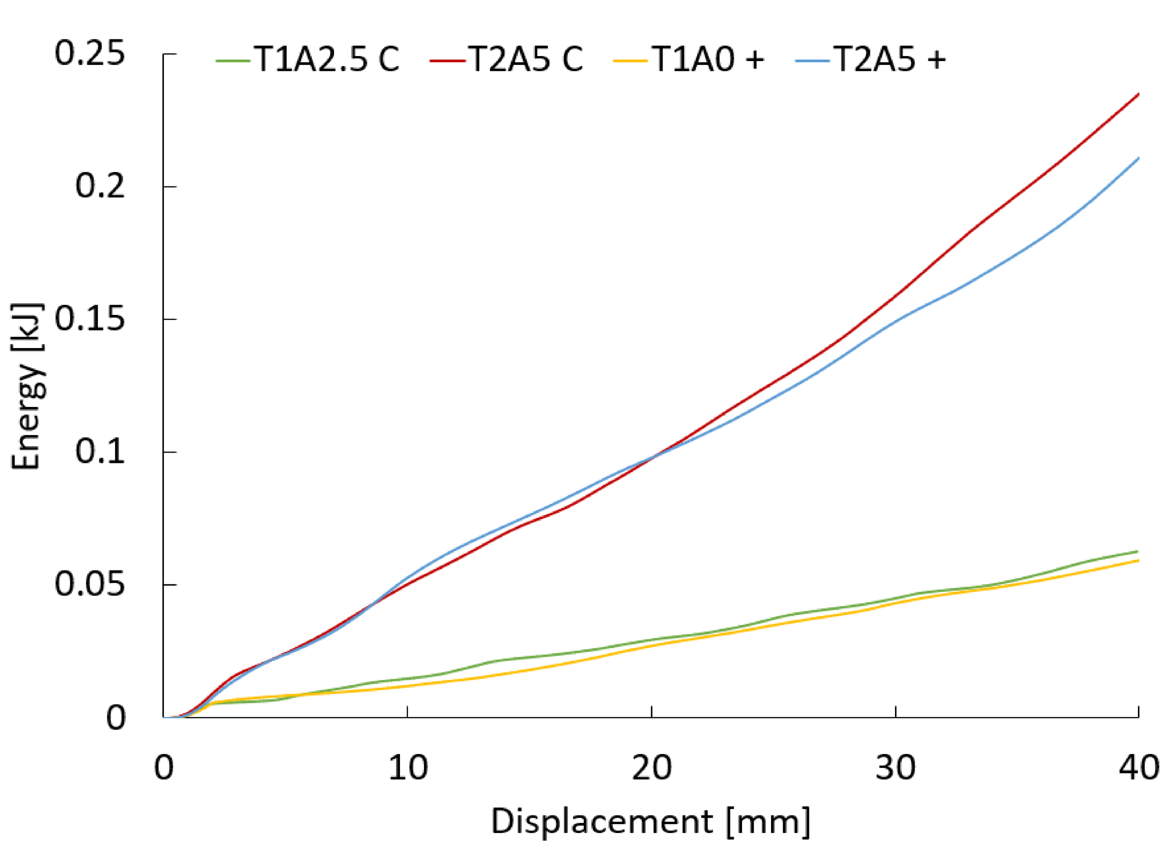
Case	Mass [g]	Energy [kJ]	PCF [kN]	MCF [kN]	SEA [kJ/kg]	CFE
T2.5A5 C	12.1	0.33	10.92	8.34	27.57	0.76
T2.5A7.5 C	10.7	0.31	10.45	7.72	28.86	0.74
T2A7.5 C	8.8	0.22	7.31	5.51	25.04	0.75
T3A5 C	14.2	0.45	14.2	11.14	31.39	0.78
T3A7.5 C	12.5	0.39	12.17	9.85	31.53	0.81
T2.5A5 +	12.2	0.3	10.13	7.53	24.69	0.74
T2.5A7.5 +	10.8	0.31	11.3	7.69	28.47	0.68
T2A7.5 +	8.8	0.2	6.47	5.11	23.23	0.79
T3A5 +	14.2	0.42	14.97	10.62	29.92	0.71
T3A7.5 +	12.6	0.38	12.64	9.58	30.4	0.76











CONCLUSIONS AND FUTURE WORK

In this study, the crashworthiness performance of 3D-printed tapered tube crash boxes were investigated. The specimens were fabricated using Fused Deposition Modeling (FDM) with two advanced filament materials: PLA-CF and PLA+. A series of nine geometric configurations were designed by varying wall thicknesses and taper angles, and each configuration was printed using both materials to comprehensively evaluate their structural response under quasi-static axial compression. The results demonstrate that greater wall thickness and larger taper angles significantly improve energy absorption characteristics. Across all configurations, PLA-CF consistently outperformed PLA+ in both and , indicating its suitability for energy-dissipating applications. Based on the experimental dataset, second-order polynomial regression models were developed to predict and , yielding a maximum absolute percentage error of 4.97%. These models were further utilized to guide the design of new configurations. The optimal design, defined by a thickness of 3 mm and a taper angle of 7.5 degrees using PLA-CF, achieved an of 31.53 kJ/kg and a of 0.81. The findings highlight the effectiveness of tapered tube designs and the use of PLA-CF material in improving crashworthiness performance.

Reference

Tunay, M., & Bardakci, A. (2024). A study of crashworthiness performance in thin-walled multi-cell tubes 3D-printed from different polymers. *Journal of Applied Polymer Science.* https://doi.org/10.1002/app.56287

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